

CANBY, MN, USA

WMO: 722252

Lat: 44.729N

Lon: 96.266W

Elev: 364

StdP: 97.03

Time Zone: -6.00 (NAC)

Period: 06-19

WBAN: 54923

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.6	-21.7	-27.4	0.3	-23.2	-25.4	0.4	-21.4	13.3	-5.0	12.0	-7.8	2.8	290	0.575

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.7	32.8	22.6	31.3	22.1	29.9	21.4	25.2	30.5	24.0	29.1	22.9	27.7	4.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.5	19.1	28.6	22.4	17.9	27.6	21.3	16.7	26.2	78.9	30.5	73.9	28.8	69.6	27.7	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.5	9.1	-28.1	36.0	2.8	2.0	-30.1	37.5	-31.8	38.7	-33.4	39.8	-35.5	41.3		
(5)			-28.3	26.6	2.8	1.0	-30.3	27.3	-32.0	27.9	-33.5	28.4	-35.6	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-8.5	-7.3	0.7	7.1	14.9	20.4	23.1	21.4	17.6	9.8	2.1	-5.9		
	DBStd	12.55	7.84	7.22	7.69	5.82	5.01	3.57	3.10	3.11	4.53	5.01	6.56	6.95		
	HDD10.0	2299	574	483	300	119	13	0	0	0	2	66	248	493		
	HDD18.3	4168	833	717	547	338	130	19	3	7	67	268	488	751		
	CDD10.0	1582	0	0	12	32	166	313	404	354	231	59	11	0		
	CDD18.3	409	0	0	1	1	24	82	150	103	46	3	0	0		
	CDH23.3	3983	0	0	6	28	282	777	1499	901	448	41	1	0		
	CDH26.7	1342	0	0	1	6	89	259	548	289	141	9	0	0		
(14) Wind	WSAvg	3.9	4.4	4.5	4.3	4.6	4.2	3.5	2.9	2.8	3.5	4.2	4.3	4.2		
(15) Precipitation	PrecAvg	687	19	19	37	69	83	120	85	77	76	62	33	19		
	PrecMax	928	65	41	93	182	145	247	144	141	242	194	90	56		
	PrecMin	544	1	2	11	19	26	30	25	30	16	7	1	2		
	PrecStd	102	15	11	21	37	38	57	37	30	53	48	29	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.1	14.3	23.5	26.6	32.5	34.3	35.1	33.6	32.7	28.0	21.5	11.6		
		MCWB	4.3	10.0	14.2	14.5	18.6	23.9	23.6	23.7	21.1	17.2	13.1	7.4		
	2%	DB	6.0	8.2	18.5	23.0	29.0	32.0	32.9	31.7	30.5	23.8	17.7	8.0		
		MCWB	3.1	4.5	12.3	13.3	17.6	21.7	24.1	23.4	20.4	15.0	10.6	4.3		
	5%	DB	3.9	5.2	15.0	19.8	26.7	29.7	31.2	30.0	28.1	21.0	14.2	5.2		
		MCWB	1.9	2.9	10.3	10.9	16.9	21.3	22.7	22.2	19.6	13.7	8.8	2.4		
	10%	DB	2.2	2.8	11.6	17.0	23.8	27.7	29.8	27.9	25.9	18.0	11.4	2.9		
		MCWB	0.9	1.1	8.2	10.0	15.5	19.7	21.9	21.0	18.8	12.0	7.1	1.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.8	10.4	16.5	16.7	21.1	25.5	27.1	25.9	23.9	19.1	14.5	8.2		
		MCDB	8.5	14.4	21.5	24.6	27.7	31.7	31.4	30.8	29.1	24.3	20.0	11.1		
	2%	WB	3.4	5.0	13.2	14.5	19.5	23.9	25.5	24.5	22.6	16.3	11.4	4.4		
		MCDB	5.8	7.6	17.5	19.7	26.0	29.0	30.5	29.9	27.3	21.8	16.0	7.4		
	5%	WB	2.1	2.9	10.5	12.8	18.2	22.3	24.4	23.5	21.4	14.5	9.3	2.8		
		MCDB	3.7	5.1	14.8	17.9	24.2	27.7	29.6	28.2	26.0	19.7	14.1	4.7		
	10%	WB	0.9	1.2	7.8	10.8	16.9	21.0	23.2	22.2	20.0	12.7	7.2	1.5		
		MCDB	2.2	2.7	11.9	16.2	22.2	26.1	28.1	26.4	24.6	17.3	11.1	3.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.3	9.4	9.8	11.0	11.5	11.1	11.7	11.2	12.3	10.5	9.8	8.3		
		MCDBR	9.9	11.3	14.6	17.5	15.1	13.9	12.7	13.0	14.7	14.7	13.6	10.4		
	5% WB	MCWBR	7.4	8.3	9.0	8.9	6.8	6.6	5.8	6.1	6.8	7.9	7.7	7.5		
		MCWBR	9.3	10.9	13.8	13.9	12.5	11.9	11.5	11.6	12.1	12.8	12.8	9.3		
(40) Clear-Sky Solar Irradiance	taub		0.265	0.281	0.315	0.360	0.377	0.389	0.397	0.402	0.359	0.322	0.286	0.269		
	taud		2.443	2.413	2.412	2.327	2.351	2.375	2.345	2.307	2.410	2.490	2.507	2.444		
	Ebn at Noon		869	916	923	901	893	880	867	849	862	854	837	824		
	Edh at Noon		70	87	101	120	121	119	121	121	101	81	66	64		
(44) All-Sky Solar Radiation	RadAvg		1.83	2.88	3.79	4.77	5.52	6.14	6.61	5.46	4.26	2.79	1.84	1.37		
	RadStd		0.20	0.32	0.38	0.45	0.41	0.41	0.24	0.46	0.32	0.36	0.17	0.13		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page